October, 2020

B.Tech. (CE/CSE/IT)-IV SEMESTER

Design & Analysis of Algorithms (PCC-CS-404)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*
2. *Answer any four questions from Part -B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART - A

1. (a) Write the recurrence relation for ternary search and also solve it. (1.5)

   (b) Differentiate big 'O' and small 'o' asymptotic notations. (1.5)

   (c) If an array is sorted in decreasing order then which sorting runs in minimal complexity? Justify. (1.5)

   (d) Explain Transitive Closure in graph. (1.5)

   (e) Differentiate Binary tree, Binary Search Tree and Optimal Binary Search Tree. (1.5)

(f) Merge the following files optimally :

(1, 3, 2, 5, 3, 4, 6, and 8)

(g) Describe principle of optimality. (1.5)

(h) Explain explicit and implicit constraints for Hamiltonian cycle. (1.5)

(i) Describe Least Cost Search function in brief. (1.5)

(j) Differentiate NP-Hard and NP-Complete problems. (1.5)

PART - B

2. (a) Solve the following recurrence relations:-

(i) $T(n) = T(n-1) + n$

(ii) $T(n) = T(\sqrt{n}) + 1$

(iii) $T(n) = 3T(n/9) + n^3$

(iv) $T(n) = T(n/3) + T(2n/3) + n$ (10)

(b) Solve the knapsack problem (0/1 and Fractional) using Greedy Method: m = 30, (w1, w2, w3, w4) = (10, 15, 6, 9), (p1, p2, p3, p4) = (2, 5, 8, 1). (5)

3. (a) Define Merge-Purge rule with an appropriate example. (5)

(b) Write the algorithm for Quick-sort and compute its time complexity. Also sort the following array using Quick-sort:- (10)

{10, 23, 6, 34, 21, 76}

4. Solve the following travelling salesman problem using branch and bound:

0	7	3	12	8
3	0	6	14	9
5	8	0	6	18
9	3	5	0	11
18	14	9	8	0

(15)

5. (a) Write the backtracking algorithm for n-queen problem. Find a solution to place 4 queens on a 4*4 chess-board. (5)

(b) Write Network Flow algorithm and explain with appropriate example. (10)

6. (a) Explain Approximation algorithms in detail. (10)

(b) Define Strassen's Matrix Multiplication in brief. (5)

7. Define Cook's theorem and explain reduction of NP-Hard problems into NP-Complete problem with suitable example. (15)

Sr. No. 003404

May 2024

B.Tech(CE/CE(HINDI)/IT/CSE(AIML)) - IV SEMESTER

Design & Analysis of Algorithms (PCC-CS-404)

एल्गोरिदम का डिजाईन और विश्लेषण (पीसीसी-सीएस-404)

Time(समय): 3 Hours(3 घंटे) Max. Marks(कुल अंक) :75

Instructions: 1. *It is compulsory to answer all the questions (1.5 marks each) of Part -A in short.*

निर्देश: भाग-क के सभी प्रश्नों (प्रत्येक 1.5 अंक) का उत्तर संक्षेप में देना अनिवार्य है

2. *Answer any four questions from Part -B in detail.*

भाग-ख से किन्हीं चार प्रश्नों के उत्तर विस्तार से दीजिए।

3. *Different sub-parts of a question are to be attempted adjacent to each other.*

एक प्रश्न के विभिन्न उप-भागों को एक दूसरे के निकट करने का प्रयास किया जाना है।

PART–A (भाग - क)

Q1 (a) Write the recurrence relation for n-ary search and why it is not preferred? (1.5)

एन-एरी खोज के लिए पुनरावृत्ति संबंध लिखें और इसे प्राथमिकता क्यों नहीं दी जाती है?

(b) Write the following functions in the increasing order of asymptotic complexity: (1.5)

निम्नलिखित कार्यों को स्पर्शोन्मुख जटिलता के बढ़ते क्रम में लिखिए?

$f1(n) = 2^n$ $f2(n) = n^{(3/2)}$ $f3(n) = n*\log(n)$ $f4(n) = n^{\log(n)}$

(c) Why Greedy method is not suggested for 0/1 Knapsack problem? (1.5)

0/1 नैपसेक समस्या के लिए लालची विधि का सुझाव क्यों नहीं दिया जाता है?

(d) Bellman-Ford Algorithm always find negative weight cycle in the graph? Justify your answer. (1.5)

बेलमैन-फोर्ड एल्गोरिथम हमेशा ग्राफ़ में नकारात्मक वजन चक्र ढूंढता है? अपने जवाब का औचित्य साबित करें।

(e) In which scenario a sorting technique is called stable? Also give example of 2 stable sorting techniques. (1.5)

किस परिदृश्य में छँटाई तकनीक को स्थिर कहा जाता है? दो स्थिर छँटाई तकनीकों का उदाहरण भी दें।

(f) Merge the following files optimally: (5,9,4,2,1,10) (1.5)

निम्नलिखित फ़ाइलों को इष्टतम रूप से मर्ज करें: (5,9,4,2,1,10)

(g) Explain the significance of asymptotic notations. (1.5)

स्पर्शोन्मुख संकेतन के महत्व को समझाइये।

(h) Explain explicit and implicit constraints for Hamiltonian Cycle. (1.5)

हैमिल्टनियन चक्र के लिए स्पष्ट और अंतर्निहित बाधाओं की व्याख्या करें।

(i) Describe Least Cost Search function in brief. (1.5)

(j) Differentiate P, NP-Hard and NP-Complete problems. (1.5)

पी, एनपी-हार्ड और एनपी-कम्प्लीट समस्याओं में अंतर करें।

PART-B (भाग - ख)

Q2 (a) Solve the following recurrence relations: (5)

निम्नलिखित पुनरावृत्ति संबंधों को हल करें:

(i) $T(n) = T(\sqrt{n}) + n$

(ii) $T(n) = 7T\ \text{Floor}(n/3) + n^2$

(b) Write the algorithm for divide and conquer based sorting technique which runs with same time complexity in all scenarios and also compute the time complexity of the algorithm. (10)

डिवाइड और कॉनकॉर पर आधारित सॉर्टिंग तकनीक के लिए एल्गोरिदम लिखें जो सभी परिदृश्यों में समान समय जटिलता के साथ चलता है और एल्गोरिदम की समय जटिलता की गणना भी करता है।

Q3 (a) Explain dominance rule (Merge-Purge) with an appropriate example. (5)

प्रभुत्व नियम (मर्ज-पर्ज) को उचित उदाहरण सहित समझाइये।

(b) Differentiate backtracking and dynamic programming algorithm design techniques corresponding to their application domain. Also explain the N-Queen problem in detail using backtracking. (10)

उनके एप्लिकेशन डोमेन के अनुरूप बैकट्रैकिंग और डायनेमिक प्रोग्रामिंग एल्गोरिदम डिज़ाइन तकनीकों को अलग करें। बैकट्रैकिंग का उपयोग करके एन-क्वीन समस्या को भी विस्तार से समझाएं।

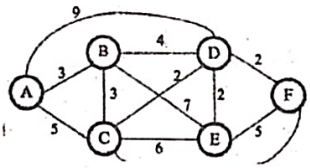
Q4 (a) Amit resides at city A and plans to visit various cities B, C and D. Compute optimal route with minimum cost to return back to A and every city he wants to visit once. (15)

अमित शहर A में रहता है और विभिन्न शहरों B, C और D का दौरा करने की योजना बना रहा है। A और प्रत्येक शहर में वापस लौटने के लिए न्यूनतम लागत के साथ इष्टतम मार्ग की गणना करें, जहां वह एक बार जाना चाहता है।

	A	B	C	D
A	0	8	4	12
B	4	0	6	14
C	6	8	0	7
D	9	4	5	0

Q5 (a) Compute the Shortest path from A to F using greedy method. (5)

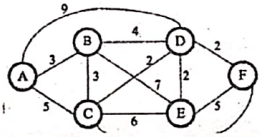
लालची पद्धति का उपयोग करके A से F तक के सबसे छोटे पथ की गणना करें।



	A	B	C	D
A	0	8	4	12
B	4	0	6	14
C	6	8	0	7
D	9	4	5	0

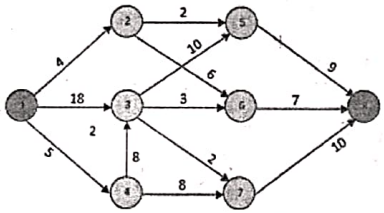
Q5 (a) Compute the Shortest path from A to F using greedy method. (5)

लालची पद्धति का उपयोग करके A से F तक के सबसे छोटे पथ की गणना करें।



(b) Write Ford Fulkerson-Network Flow algorithm and compute the maximum unit flow from node 1 to node 8. (10)

फोर्ड फुलकर्सन-नेटवर्क फ्लो एल्गोरिदम लिखें और नोड 1 से नोड 8 तक अधिकतम इकाई प्रवाह की गणना करें।



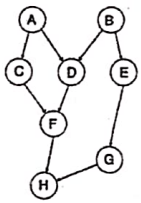
Q6 (a) Solve the knapsack problem using Branch and bound technique: (10)

ब्रांच और बाउंड तकनीक का उपयोग करके नैपसैक समस्या को हल करें

m=30, (w1, w2, w3, w4) = (18, 15, 8, 12), (p1, p2, p3, p4) = (36, 24, 8, 12)

(b) If A to H are different steps of a task and their dependency is given in the graph below. Compute topological order of events for successfully completion of task. (5)

यदि A से H किसी कार्य के अलग-अलग चरण हैं और उनकी निर्भरता नीचे ग्राफ़ में दी गई है। कार्य को सफलतापूर्वक पूरा करने के लिए घटनाओं के टोपोलॉजिकल क्रम की गणना करें।



Q7 (a) Explain any 2 types of Randomized algorithm techniques with corresponding applications and scope. (8)

संबंधित अनुप्रयोगों और दायरे के साथ किन्हीं 2 प्रकार की यादृच्छिक एल्गोरिदम तकनीकों की व्याख्या करें।

(b) Explain the utility of Approximation algorithms with appropriate example. (7)

उचित उदाहरण सहित एप्रोक्सिमेशन एल्गोरिदम की उपयोगिता समझाइए

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May 2023

B.Tech. - IV SEMESTER

Design & Analysis of Algorithms (PCC-CS-404)

Time : 3 Hours] [Max. Marks : 75

Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) What is the time complexity of the following code? Justify your answer.

```
int i, j, k = 0;
  for (i = n / 2; i <= n; i++) {
    for (j = 2; j <= n; j = j * 2) {
      k = k + n/2;
    }}
```

(1.5)

(b) Sort the following functions in the decreasing order of their asymptotic (big-0) complexity : $f_1(n) = n^{\sqrt{n}}$, $f_2(n) = 2^n$, $f_3(n) = (1.000001)^n$, $f_4(n) = n^{10} * 2^{(n/2)}$. (1.5)

(c) Differentiate Greedy algorithm and Dynamic programming. (1.5)

(d) State Job Sequencing with Deadline Problem. Write down time and space complexity if problem solved by Greedy approach. (1.5)

(e) Define Principle of Optimality with suitable example. (1.5)

(f) Draw state space tree of 4-Queens problem. (1.5)

(g) Why does Dijkstra's algorithm fail on negative weights? (1.5)

(h) Draw binary search trees for the given set of keys and their corresponding frequencies and find the Optimal among them. keys[ ] = {10, 12, 20}, freq[ ] = {34, 8, 50} (1.5)

(i) Explain Least Cost Search function for branch and bound algorithm design technique. (1.5)

(j) What is the importance of approximation algorithm? (1.5)

PART–B

2. (a) Solve the following recurrence relation.

(i) $T(n) = 2T(n/2) + \log n$ with $T(1)=1$

(ii) $T(n) = 2T(\sqrt{n}) + 1$ if $n > 2$ and $T(n) = 2$ if $0<n<=2$. (10)

(b) How are time and space trade-offs used to optimize the performance of an algorithm? Provide an example of an algorithm that optimizes time at the expense of space and vice-versa. (5)

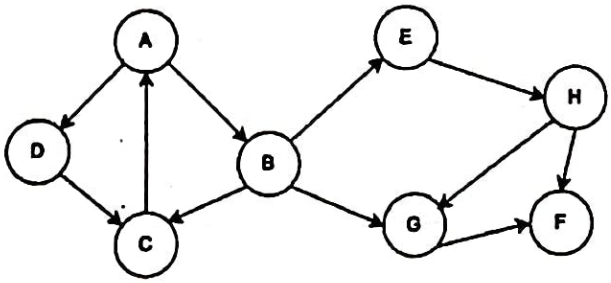
3. (a) Give a dynamic-programming solution to the 0-1 knapsack problem that runs in 0(n/W) time, where n is the number of items and W is the maximum weight of items that the thief can put in his knapsack.

Consider the weights and values of items listed below. The task is to pick a subset of these items such that their total weight is no more than 5 Kgs and o their total value is rnaximized.

Item No.	Weight (Kg)	Values (Rs.)
1	2	3
2	3	4
3	4	5
4	5	6

(b) Consider the given graph. In what order will the nodes be visited using a Breadth First Search and Depth First Search? (Assume starting vertex A) (5)

Note: If a node has multiple neighbors then select which is alphabetically near to node.



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May, 2019

B.TECH. (CE/CSE/IT) - IV SEMESTER

Design & Analysis of Algorithms (PCC-CS-404)

Time : 3 Hours]

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Instructions :

1. *It is compulsory to answer all the questions (1.5 marks each) of Part-A in short.*
2. *Answer any four questions from Part-B in detail.*
3. *Different sub-parts of a question are to be attempted adjacent to each other.*

PART–A

1. (a) Solve the following recurrence relation:

$$T(n) = T(n^{1/2}) + c.$$ (1.5)

(b) Differentiate between O-notation and Omega-notation with the help of appropriate example. (1.5)

(c) How arrays can be used to store a tree? Discuss with example. (1.5)

(d) For which type of problems Divide & Conquer approach is generally used, discuss. (1.5)

(e) Differentiate between Depth First Search and Breadth First Search. (1.5)

(f) What is Transitive Closure of a graph? (1.5)

(g) What do you mean by reducibility? (1.5)

(h) Which approach among Backtracking and Branch & Bound is better and why? (1.5)

(i) Discuss the time complexity of Kruskal's algorithm. (1.5)

(j) Why Randomized algorithms are used? Discuss with example. (1.5)

PART–B

2. Design a divide and conquer algorithm to find the maximum and minimum of an array A of n elements, and prove that the algorithm makes at most 3n/2 element-to-element comparisons. (15)

3. (a) Differentiate between Deterministic and Non-Deterministic algorithms. Write a Deterministic algorithm to search an element in the array and convert the same into a Non-Deterministic algorithm. (7)

(b) Differentiate between Greedy and Dynamic method to solve the problems. Write and explain All Pair Shortest Path algorithm to find the Shortest Paths in a graph with example and derive its time complexity. (8)

4. Find all the feasible solutions to the Knapsack instance $n = 7$, $m = 15$,

$(P_i) = (10, 5, 15, 7, 6, 18, 3)$

$(W_i) = (2, 3, 5, 7, 1, 4, 1)$.

Find the optimal solution for the same by using Greedy method. (15)

5. (a) What do you mean by Network Flow? Write Ford-Fulkerson algorithm with suitable example. (5)

(b) Differentiate between Backtracking and Branch & Bound with the help of suitable examples. Discuss the Hamiltonian Cycle problem and write its algorithm. (10)

6. (a) Why Topological sorting is used for Directed Acyclic Graph (DAG), justify with suitable example. (7)

(b) What are approximation algorithm and why they are used? Solve Travelling Salesman Problem by using approximation algorithm. (5)

7. (a) Discuss P, NP, NP-Complete and NP-Hard Problems with suitable examples. (10)

(b) What is Satisfibility? Discuss CNF-Satisfibility. (5)